PRACTICAL NO. 2

FITTING SHOP

Introduction

Machine tools are capable of producing work at a faster rate, but, there are occasions when components are processed at the bench. Sometimes, it becomes necessary to replace or repair component which must be fit accurately with another component on reassembly. This involves a certain amount of hand fitting. The assembly of machine tools, jigs, gauges, etc. involves certain amount of bench work. The accuracy of work done depends upon the experience and skill of the fitter.

The term 'bench work' refers to the production of components by hand on the bench, whereas fitting deals which the assembly of mating parts, through removal of metal, to obtain the required fit.

Both the bench work and fitting requires the use of number of simple hand tools and considerable manual efforts. The operations in the above works consist of filing, chipping, scraping, sawing drilling, and tapping.

Holding Tools:

Bench Vice

The bench vice is a work holding device. It is the most commonly used vice in a fitting shop.

The bench vice is shown in figure below.

It is fixed to the bench with bolts and nuts. The vice body consists of two main parts, fixed jaw and movable jaw. When the vice handle is turned in a clockwise direction, the sliding jaw forces the work against the fixed jaw. Jaw plates are made of hardened steel. Serrations on the jaws ensure a good grip. Jaw caps made of soft material are used to protect finished surfaces, gripped in the vice. The size of the vice is specified by the length of the jaws.

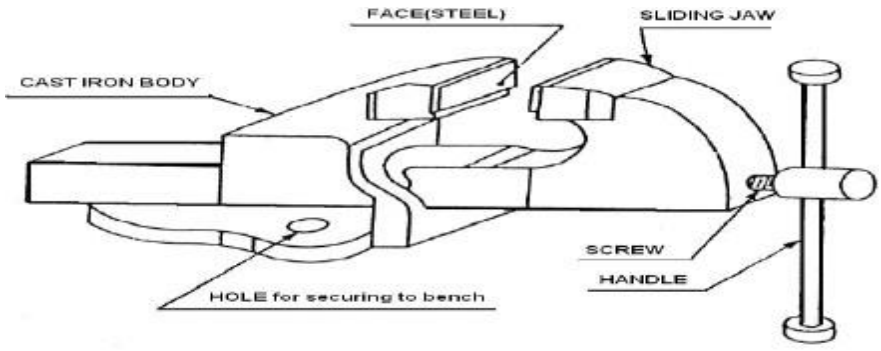


Fig: Bench Vice

The vice body is made of cast iron which is strong in compression, weak in tension and so fractures under shocks and therefore should never be hammered. V-block is rectangular or square block with a V-groove on one or both sides opposite to each other. The angle of the 'V' is usually 900. V-block with a clamp is used to hold cylindrical work securely, during layout of measurement, for measuring operations or for drilling for this the bar is faced longitudinally in the V-Groove and the screw of V-clamp is tightened. This grip the rod is firm with its axis parallel to the axis of the v-groove.

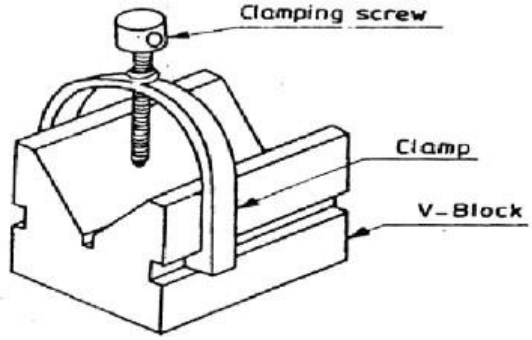


Fig: V- Block

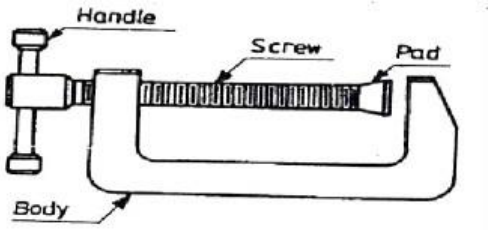


Fig: C - Clamp

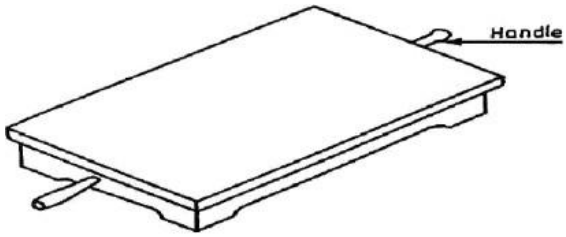
C- Clamp

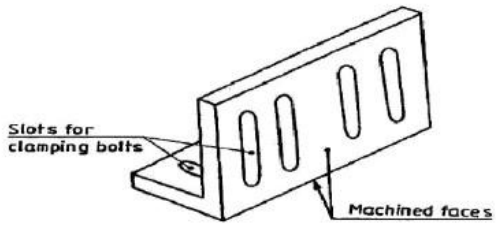
This is used to hold work against an angle plate or v-block or any other surface, when gripping is require Its fixed jaw is shaped like English alphabet 'C' and the movable jaw is round in shape and directly fitted to the threaded screw at the end . The working principle of this clamp is the same as that of the bench vice.

Marking and Measuring Tools:

Surface Plate

The surface plate is machined to fine limits and is used for testing the flatness of the work piece. It is also used for marking out small box and is more precious than the marking table. The degree of the finished depends upon whether it is designed for bench work in a fitting shop or for using in an inspection room; the surface plate is made of Cast Iron, hardened Steel or Granite stone. It is specified by length, width, height and grade. Handles are provided on two opposite sides, to carry it while shifting from one place to another (refer figure in next page).



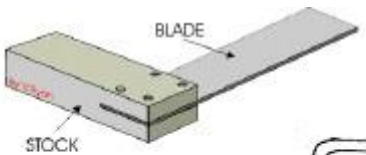


It is measuring and marking tool for 900 angle .In practice, it is used for checking the squareness of many types of small works when extreme accuracy is not required .The blade of the Try square is made of hardened steel and the stock of cast Iron or steel. The size of the Try square is specified by the length of the blade.

Scriber

A Scriber is a slender steel tool, used to scribe or mark lines on metal work pieces. It is made of hardened and tempered High Carbon Steel. The Tip of the scriber is generally ground at 12° to 15°.

It is generally available in lengths, ranging from 125mm to 250mm .It has two pointed ends the bent end is used for marking lines where the straight end cannot real**.**



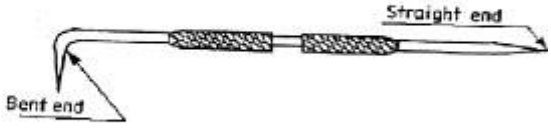


Fig: Try Square **Fig: Scriber**

Odd Leg Caliper

This is also called 'Jenny Caliper' or Hermaphrodite. This is used for marking parallel liners from a finished edge and also for locating the center of round bars; it has one leg pointed like a divider and the other leg bent like a caliper. It is specified by the length of the leg up to the hinge point.

Divider

It is basically similar to the calipers except that its legs are kept straight and pointed at the measuring edge. This is used for marking circles, arcs laying out perpendicular lines, by setting lines. It is made of case hardened mild steel or hardened and tempered low carbon steel. Its size is specified by the length of the leg**.**

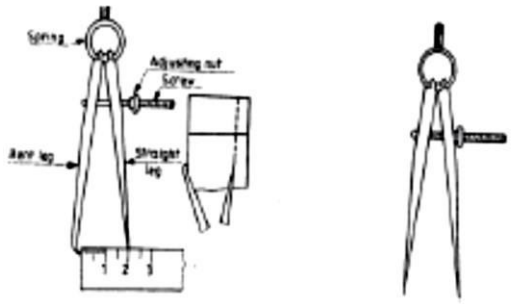


Fig: Odd Leg Caliper and Divider

Trammel is used for drawing large circles or arcs.

Punches

These are used for making indentations on the scribed lines, to make them visible clearly. These are made of high carbon steel. A punch is specified by its length and diameter (say as 150’ 12.5mm). It consists of a cylindrical knurled body, which is plain for some length at the top of it. At the other end, it is ground to a point. The tapered point of the punch is hardened over a length of 20 to 30 mm.

Dot Punch is used to lightly indent along the layout lines, to locate center of holes and to provide a small center mark for divider point, etc. for this purpose, the punch is ground to a conical point having 60° included angle.

Center Punch is similar to the dot punch, except that it is ground to a conical point having 90° included angle. It is used to mark the location of the holes to be drilled.

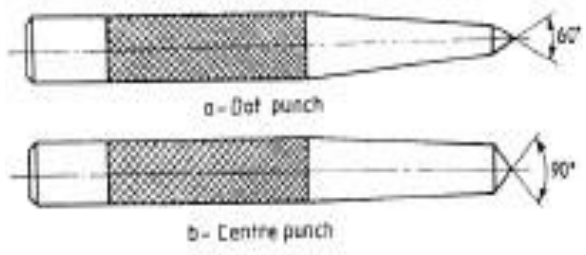


Fig: Punches

Calipers:

They are indirect measuring tools used to measure or transfer linear dimensions. These are used with the help of a steel Rule to check inside and outside measurements. These are made of Case hardened mild steel or hardened and tempered low carbon steel. While using, but the legs of the caliper are set against the surface of the work, whether inside or outside and the distance

between the legs is measured with the help of a scale and the same can be transferred to another desired place. These are specified by the length of the leg. In the case of outside caliper, the legs are bent inwards and in the case of inside caliper, the legs bent outward.

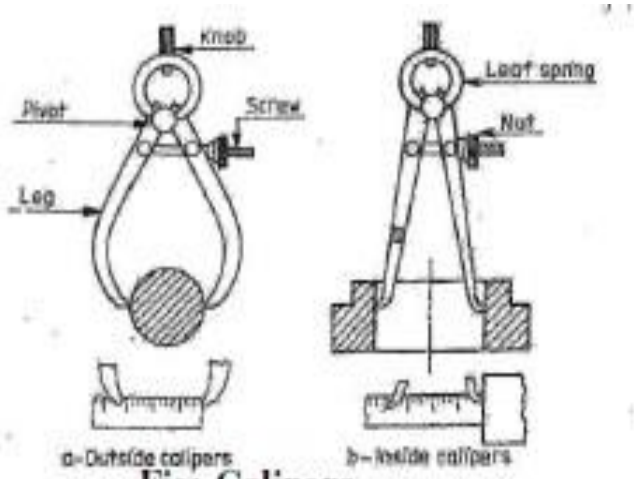


Fig: Calipers

Cutting Tools:

Hack Saw

The Hack Saw is used for cutting metal by hand. It consists of a frame, which holds a thin blade, firmly in position. Hacksaw blade is specified by the number of teeth for centimeter. Hacksaw blades have a number of teeth ranging from 5 to 15 per centimeter (cm). Blades having lesser number of teeth per cm are used for cutting soft materials like aluminum, brass and bronze. Blades having larger number of teeth per centimeter are used for cutting hard materials like steel and cast Iron.

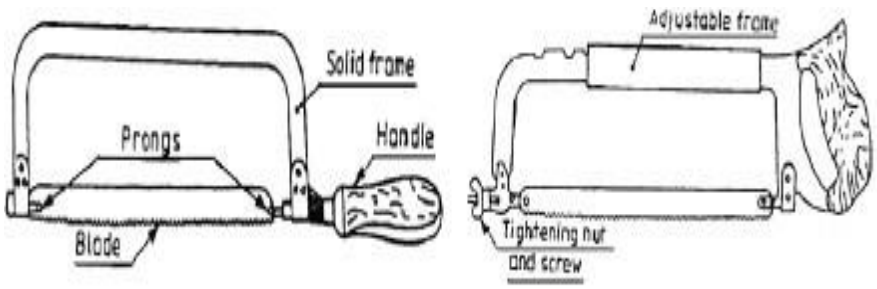


Fig: Hacksaw Frame with Blade

Hacksaw blades are classified as (i) All hard and (ii) flexible type. The all hard blades are made of H.S.S, hardened and tempered throughout to retain their cutting edges longer. These are used to cut hard metals. These blades are hard and brittle and can break easily by twisting and forcing them into the work while sawing. Flexible blades are made of H.S.S or low alloy steel but only

the teeth are hardened and the rest of the blade is soft and flexible. These are suitable for use by un-skilled or semi-skilled persons. The teeth of the hacksaw blade are staggered, as shown in figure and known as a 'set of teeth'.
These make slots wider than the blade thickness, preventing the blade from jamming.

Chisels

Chisels are used for removing surplus metal or for cutting thin sheets. These tools are made from 0.9% to 1.0% carbon steel of octagonal or hexagonal section. Chisels are annealed, hardened and tempered to produce a tough shank and hard cutting edge. Annealing relieves the internal stresses in a metal. The cutting angle of the chisel for general purpose is about 60°.

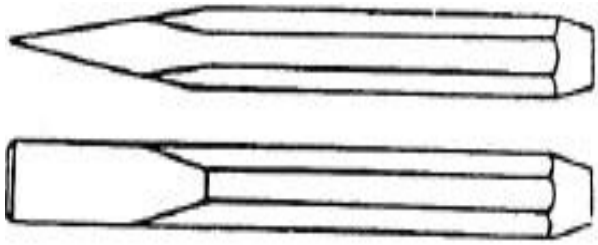


Fig: Flat Chisel

Twist Drill

Twist drills are used for making holes. These are made of High speed steel. Both straight and taper shank twist drills are used. The parallel shank twist drill can be held in an ordinary self – centering drill check. The tapper shank twist drill fits into a corresponding tapered bore providedin the drilling machine spindle (see figure in next page).

Taps and Tap Wrenches

A tap is a hardened and steel tool, used for cutting internal thread in a drill hole. Hand Taps are usually supplied in sets of three in each diameter and thread size. Each set consists of a tapper tap, intermediate tap and plug or bottoming tap. Taps are made of high carbon steel or high speed steel (see figure in next page).

Bench Drilling Machine

Holes are drilled for fastening parts with rivets, bolts or for producing internal thread. Bench drilling machine is the most versatile machine used in a fitting shop for the purpose. Twist drills, made of tool steel or high speed steel are used with the drilling machine for drilling holes.

Following are the stages in drilling work
1. Select the correct size drills, put it into the check and lock it firmly

2. Adjust the speed of the machine to suit the work by changing the belt on the pulleys. Use high speed for small drills and soft materials and low speed for large diameter drills and hard materials.

3. Layout of the location of the pole and mark it with a center punch.

4. Hold the work firmly in the vice on the machine table and clamp it directly on to the machine table.

5. Put on the power, locate the punch mark and apply slight pressure with the Feed Handle.

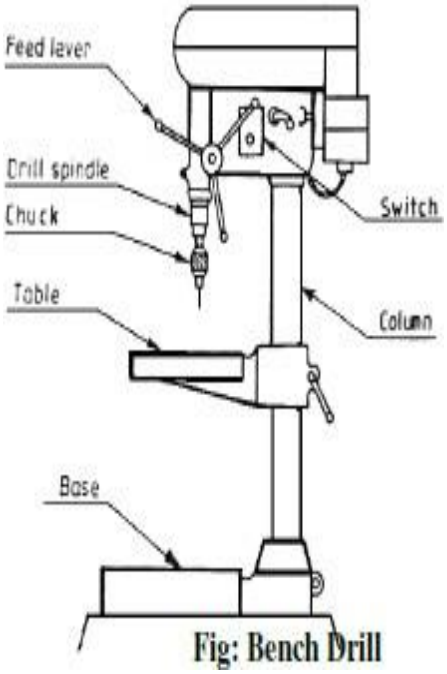


Fig: Bench Drill

Finishing Tools:

Reamers

Reaming is an operation of sizing and finishing a drilled hole, with the help of a cutting tool called reamer having a number of cutting edges. For this, a hole is first drilled, the size of which is slightly smaller than the finished size and then a hand reamer or machine reamer is used for finishing the hole to the correct size. Hand Reamer is made of High Carbon Steel and has left-hand spiral flutes so that, it is prevented from screwing into the whole during operation. The Shank end of the reamer is made straight so that it can be held in a tap wrench. It is operated by hand, with a tap wrench fitted on the square end of the reamer and with the work piece held in the vice. The body of the reamer is given a slight tapper at its working end, for its easy entry into the whole during operation, it is rotated only in clock wise direction and also while removing it from the whole.

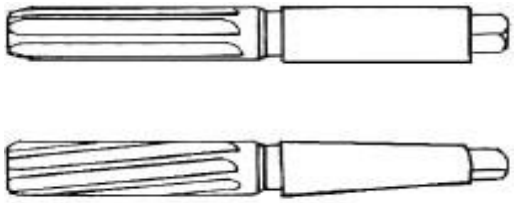


Fig: Reamers

Files

Filing is one of the methods of removing small amounts of material from the surface of a metal part. A file is hardened steel too, having small parallel rows of cutting edges or teeth on its surfaces.

On the faces, the teeth are usually diagonal to the edge. One end of the file is shaped to fit into a wooden handle. The figure shows various parts of a hand file. The hand file is parallel in width and tapering slightly in thickness, towards the tip. It is provided with double cut teeth. On the faces, single cut on one edge and no teeth on the other edge, which is known as a safe edge.

Fig: Parts of a Hand File

Files are

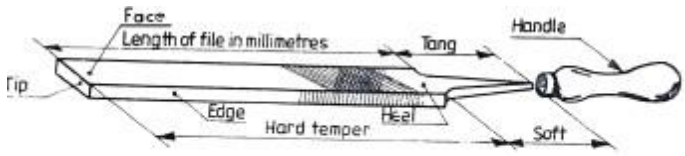


Fig: Parts of a Hand File

Files are classified according to their shape, cutting teeth and pitch or grade of the teeth. The figure shows the various types of files based on their shape.

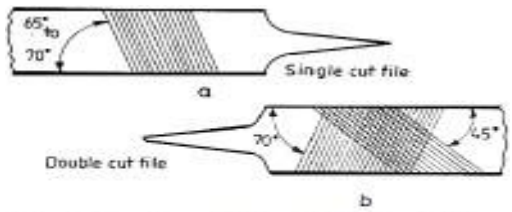


Fig: Single and Double Cut Files

Miscellaneous Tools:

Ball- Peen Hammer

Ball- Peen Hammers are named, depending upon their shape and material and specified by their weight. A ball peen hammer has a flat face which is used for general work and a ball end, particularly used for riveting.

JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
An Autonomous Institute, with NAAC "A" Grade

Session: 2023-24
FIRST YEAR SEM-I (Regular)
B. Tech. Branch (AI, CSE-A,CSE-B, DS)

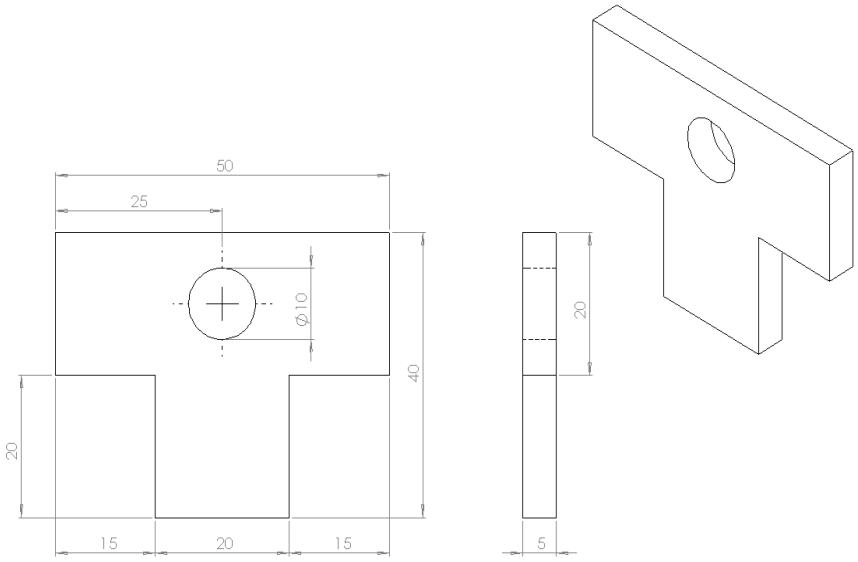
Job Drawing – Fitting Shop

Name of the job: To make a T plug on M.S. plate

Require Tools and Equipments: Carpentry Shop , Hacksaw frame with blade, Ball peen Hammer, Center punch, Scriber, Tri-Square, Chalk, Steel rule, Drill Machine, Flat file etc.

Raw material: M.S. plate length x width x thickness 50 x 40 x 05 – 1Nos

Job Drawing:



Note: All Dimensions are in mm **Scale: Not to Scale.**

Job Operations:

Sr. No.	Operation Name	Sr. No.	Operation Name
1	Cutting	5	Drilling
2	Filling	6	Cutting
3	Marking & Measuring	7	Finishing
4	Punching	8	

NAME OF THE SHOP : - Fitting Shop

NAME OF THE JOB : - To make a T plug on M.S. plate

MATERIAL REQUIRED: - M.S. plate length x width x thickness 50 x 40 x05 – 1 Nos

TOOLS AND EQPT.REQUIRED:- Hacksaw frame with blade, BallpeenHammer, Center punch, Scriber, Tri-Square, Chalk, Steel rule, Drill Machine, Flat file, Triangular File etc.

OPERATION SHEET:-

S.N.	OPERATION	TOOLS REQUIRED	PROCEDURE
1	Cutting	Power Hack-Saw Cut-off M/C	
2	Measuring & Marking	Vernier Heightgauge, angle plate, surface plate	
3	Punching	Center punch	
4	Drilling	Vertical drill machine drill bit (10 mm)	
5	Filing	Flat File	
6	Cutting	Hack saw frame with blade	
7	Finishing	Flat file	

Result: The T plug on M.S. plate is successfully done.